MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
FOURTH SEMESTER EXAMINATION
(2024 Admission Onwards)

MG4SECECT202 – Solar Technology and Applications

Duration: 1½ Hours **Maximum Marks:** 50

*Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Part A
Multiple Choice Questions
Answer all questions. Each question carries 1 mark

1. A solar cell works on the principle of
   A) Thermionic emission
   B) Photovoltaic effect
   C) Photoelectric effect
   D) Electromagnetic induction **U [CO 1]**
2. The _____ and _____ output of a solar cell depends on the amount of light falling on it.
   A) Current, voltage
   B) Resistance, voltage
   C) Current, resistance
   D) Resistance, inductance U [CO 1]
3. A photon is best defined as
   A) A positively charged subatomic particle
   B) A continuous electromagnetic wave
   C) A discrete packet (quantum) of light energy
   D) An instrument used to measure light intensity **U [CO 1]**
4. The basic structure of a PV cell involves a
   A) P-I-N junction
   B) P-P junction
   C) N-N junction
   D) P-N junction **U [CO 1]**
5. In a grid connected system, power is fed to the grid during…….
   A) Day time
   B) Evening time
   C) Sunset
   D) None of these **U [CO 3]**
6. A solar water pumping system utilizes energy from ……. to pump water.
   A) The three phase electric line
   B) The battery

C) The sun
D) The moon **U [CO 3]**

7. ……. is a well-known metric standard tool for measuring wire diameter.
   A) WWG
   B) SWG
   C) SSG
   D) WSG U **[CO 2]**
8. The current ministry that has the role of enforcing Renewable Energy Laws In India for solar energy is the ……..
   A) Ministry of Petroleum and Natural Resources (MPNR)
   B) Ministry of Energy (ME)
   C) Ministry of New and Renewable Energy (MNRE)
   D) Ministry of Alternate Source of Energy (MASE) U [CO 2]
9. Solar cells are most commonly made from
   A) Silver
   B) Copper
   C) Silicon **U [CO 1]**
10. Solar cell efficiency depends on
    A) Light intensity
    B) Semiconductor material
    C) Temperature
    D) All of the above **U [CO 2]**
11. If the area of a solar cell is increased, the current generated
    A) Remains the same
    B) Decreases
    C) Increases
    D) None of these **U [CO 2]**
12. For solar PV system applications, ……. are commonly used.
    A) Copper wire
    B) Aluminium wires
    C) Plastic wires
    D) Brass wires **U [CO 2]**
13. Light falling perpendicular to PV modules gives
    A) Minimum output power
    B) Below average output power
    C) Maximum output power
    D) Above average output power **U [CO 3]**
14. If the terminal voltage of a solar cell is 0.5 V and 36 cells are connected in series, the total voltage is
    A) 0.5 V
    B) 9 V
    C) 18 V
    D) 36 V U **[CO 2]**
15. A solar array is a
    A) Cascade connected solar plant
    B) Combination of solar panel with inverter

C) Series and parallel combination of solar cells
D) Parallel combination of solar plants **U [CO 1]**

16. If two PV modules with short-circuit currents Isc_1 and Isc_2 are connected in parallel, the total current is
A) $Isc_1 + Isc_2$
B) $Isc_1 \times Isc_2$
C) $Isc_1 - Isc_2$
D) $Isc_1 \div Isc_2$ **U [CO 4]**

17. Which of the following is a safety precaution when handling PV modules?
A) Disconnect the modules while they are operating
B) Wear insulated gloves and protective gear
C) Use lenes to concentrate artificial sunlight to the module
D) Lift the module by the junction box **U [CO3]**

18. The maximum current a solar cell can produce is called
A) Open-circuit current
B) Shunt current
C) Short-circuit current
D) Null current **U [CO 4]**

19. _____ are attached to the front and back surfaces of PV cells to collect current.
A) Plastic materials
B) Insulating materials
C) Metal contacts
D) Non-metallic materials **U [CO 1]**

20. PV systems not connected to the grid are called
A) Hybrid PV systems
B) Stand-alone PV systems
C) Grid-connected PV systems
D) Central PV systems **U [CO 2]**

21. The device that stores excess electricity is
A) Inverter
B) Charge controller
C) Battery
D) Rectifier **U [CO 2]**

22. The MPPT circuit is used …… the PV modules under all conditions.
A) give enough mechanical strength
B) as a lightning protection system
C) to protected from over loading
D) to extract maximum power from **U [CO 2]**

23. An off-grid solar PV system is a solar power system …….
A) that rely on a traditional electricity grid
B) that means you are not connected in any way to your grid's power system
C) that is not connected to the electrical grid
D) that don't have a grid to use as a backup **U [CO 2]**

24. DC to AC conversion is done using
A) Charge controller
B) Rectifier

C) Inverter
D) DC-DC converter **U [CO 2]**

25. During overcharge or deep discharge, the charge controller disconnects the
A) Chopper
B) Inverter
C) Battery
D) None of these **U [CO 1]**

26. Refilling electrical energy in batteries is called
A) Discharging
B) Fill factor
C) Charging
D) Reduction process **U [CO 1]**

27. _____ solar cells form a solar panel
A) Single
B) Dual
C) Multiple
D) None of these **U [CO 1]**

28. Under STC, input power density Pin is
A) 100 W/m²
B) 1000 W/m²
C) 5000 W/m²
D) 10000 W/m² **U [CO 1]**

29. Parallel interconnection of cell strings gives
A) Low output current
B) Low output voltage
C) High output voltage
D) High output current **U [CO 1]**

30. If a 100 cm² cell produces 2 A, a 50 cm² cell produces
A) 0.5 A
B) 1 A
C) 2 A
D) 5 A U **[CO 1]**

31. Batteries connected in series result in
A) Same voltage
B) Increased voltage
C) Decreased voltage
D) No change **U [CO 1]**

32. Each half-cell of a battery consists of an electrode and
A) An anode
B) A generator
C) An electrolyte
D) A galvanic cell **U [CO 1]**

33. A solar electric vehicle is an electric vehicle powered completely or significantly by ……
A) Indirect solar energy
B) Electrochemical cells
C) Direct solar energy
D) Electric battery **U [CO 3]**

34. An example of a stand-alone PV system is
    A) Residential rooftop system
    B) Office rooftop solar system
    C) Solar water pumping system
    D) Wind-solar hybrid system **U [CO 3]**
35. Most organic photovoltaic cells are …….
    A) Molecular solar cells
    B) Polymer solar cells.
    C) Nanoparticle solar cell
    D) Plastic solar cells **U [CO 3]**

1 × 35 = 35 Marks

Part B

Short Answer Questions

Answer any THREE questions. Each question carries 5 marks

36. Discuss the basic concept of quantum dot solar cells. List its merits. **U [CO 1]**
37. Describe the factors affecting electricity generated by a solar cell. U **[CO 1]**
38. Draw the V–I characteristics of a solar cell and explain briefly. U **[CO 1]**
39. Discuss two problems that often are related to the PV solar system and how they can be solved. K **[CO 3]**
40. Draw the block diagram of a stand-alone solar PV system and explain the function of each block. K **[CO 2]**

3 × 5 = 15 Marks